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Observation of risk for concussion following 'lowering the helmet to initiate contact' penalties in the NFL

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ABSTRACT

Objective: This observational study examines the associated risk for concussion following 'lowering the helmet to initiate contact' penalties in the National Football League.

Methods: Using officially published injury data on the National Football League (NFL) website, the authors reviewed NFL Players diagnosed with a concussion from 2015 to 2022 and who received a 'lowering the helmet to initiate contact' penalty from 2018 to 2020. Reported concussions were evaluated for association with 'lowering the helmet to initiate contact' penalties. The Friedman test assessed overall differences between the years. Pairwise comparisons were performed using the Wilcoxon signed-rank test with a Bonferroni correction to adjust for multiple comparisons.

Results: The average incidence of all concussions per play was 0.45% for regular-season games. From 2018 to 2020, 86 lowering the helmet to initiate contact penalties were called. Three of 86 penalties resulted in concussions, with a 3.5% incidence. The Friedman test indicated significant differences in concussion trends across the years ($\chi^2 = 16.0$, $p = 0.0003$). Pairwise analysis with the Wilcoxon signed-rank test and adjusted for multiple comparisons with a Bonferroni correction ($\alpha = 0.05/28 = 0.0018$) showed no statistically significant differences between years. There was a significant negative correlation of overall concussion incidence per year.

Conclusions: The incidence of concussion with players penalized for lowering the helmet to initiate contact is greater than the overall incidence of concussions in all plays. While there is no difference in total concussions that occur, the concussion rate has a significant negative trend between 2015 and 2022. However, it is difficult to analyze the nonimmediate effects of this penalty on players' intended play and practice, which may further decrease concussions.

1. Introduction

There is an increasing concern for sport-related concussions (SRC) throughout athletics as more information is gleaned about their short- and long-term effects. This critical importance is due to the large and often under-reported impact on athletes. This effect is underlined by the estimate that 1.0-1.9 million SRC occur in US children under 18 alone.[1] Recently, Chronic Traumatic Encephalopathy (CTE) has been prominently presented in literature and media. CTE is progressive neurodegenerative tauopathy associated with mild head trauma, including concussions and asymptomatic sub-concussive collisions.[2] While many strides have been made to understand its pathophysiology, more must be understood, and prevention remains vital.

American football is of notable concern, accounting for one of the most frequent causes of SRC.[3] NFL has taken steps to reduce the risk of concussion, such as advances in helmet technology, alterations in gameplay, screening protocols, and penalizing players for at-risk contact.[4–6] A 2010 study commissioned by NFL's Mild Traumatic Brain Injury Committee reported a decreased incidence in compared periods of 1996–2002 and 2002–2007.[7]

In addition, the NFL Head, Neck, and Spine Committee was commissioned to develop the NFL Gameday Concussion Diagnosis and Management Protocol in 2011.[8] This protocol led to players being evaluated after impact to the head. If they expressed signs and symptoms of concussion or if medical/coaching staff deemed it necessary, players were entered into the protocol and managed clinically until game-ready.

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Table 1
NFL Preseason and Regular Season Concussions: Total number of concussions per season from NFL Injury Reports between 2015 and 2022.[13].

Year	Preseason			Regular Season			Preseason + Regular Season		
	Game	Practice	Total	Game	Practice	Total	Game	Practice	Total
2015	54	29	83	183	9	192	237	38	275
2016	45	26	71	166	6	172	211	32	243
2017	46	45	91	179	11	190	225	56	281
2018[a]	34	45	79	127	8	135	161	53	214
2019	49	30	79	136	9	145	185	39	224
2020[b]	N/A	30	30	129	13	142	129	43	172
2021[c]	22	30	52	126	9	135	148	39	187
2022[c]	26	25	51	149	12	161	176	37	213
Mean ± SD	39 ± 12.2	33 ± 7.9	67 ± 20.7	149 ± 23.7	10 ± 2.3	159 ± 23.5	184 ± 38.05	42 ± 8.3	226 ± 38.7

Data as of 23 December 2023. Injuries reported during conditioning, weightlifting, run sessions, or individual training, as well as chronic injuries or those with insidious or unknown onset, are not included.

[a] Implementation of 'Lowering the Helmet to Initiate Contact' rule.

[b] 2020 was markedly different from other seasons due to the COVID-19 pandemic. There were no preseason games in 2020.

[c] The NFL season structure changed in 2021 and 2022 to include 3 preseason games and 17 regular season games. Prior to 2021, the NFL season included 4 preseason games and 16 regular season games.

While this led to readily identifying cases of head trauma, it did not mitigate the incidence of concussions in real-time play.

Ryan Shazier attempted a tackle while lowering his helmet, resulting in a spinal cord contusion and paraplegia in 2017. In response to this injury, the NFL devised a new penalty in 2018 to eliminate lowering the head to initiate contact. The rule was implemented to remove unnecessary helmet use and decrease potentially dangerous collisions. No studies observe how often this penalty ultimately results in increasing reported concussions. This study evaluates the association between concussions and 'Lowering the Helmet to Initiate Contact' penalties. While this rule was combined with the unnecessary roughness penalty in 2020, understanding the association between concussions and the specific player position this penalty was able to isolate will help inform further safety changes. No previous studies have evaluated the concussion incidence of 'lowering the helmet' penalties to overall concussion incidence in the NFL to understand if this penalty puts players at higher risk of concussions. This study hypothesizes that 'lowering the helmet to initiate contact' penalties will be associated with a higher incidence of concussions compared to the overall incidence of concussions in the NFL.

2. Materials and methods

2.1. Study design

The inclusion criteria for this study included all NFL players who entered the concussion protocol from 2015 to 2022 and received a 'lowering the helmet' penalty from 2018 to 2020. Exclusion criteria included incomplete data on concussion diagnosis or penalty.

2.2. Data extraction

The total number of plays per season was gathered from official NFL team stats.[9] Helmet lowering to initiate contact penalties among NFL athletes reported from 2018 to 2020 were reviewed on the NFL official website (https://www.nfl.com).[10] Factors were reviewed, including player, injury, position, and the penalty's time. Following this, online game recordings and play tracking were used to identify the player who was hit by an independent researcher (G.W.) and verified by a second researcher (J.K.)

Ethical approval

All data in this study was accessed through publicly available resources from the NFL through NFL.com.[9,10] Thus, no institutional review board approval was necessary.

Table 2
Number of Plays and Incidence of Concussion per Play per Year: Total number of regular season plays from the 2015 to the 2022 National Football League Season per total number of concussions recorded from regular season games.

Year	Regular Season	
	Total Plays	Concussions/Plays (%)
2015	32973	0.55%
2016	32734	0.51%
2017	32438	0.55%
2018	32224	0.39%
2019	32516	0.42%
2020	32945	0.39%
2021[a]	34449	0.37%
2022[a]	34136	0.44%
Mean ± SD	33052 ± 758	0.45% ± 0.08%

Data as of 23 December 2023. Injuries reported during conditioning, weightlifting, run sessions, or individual training, as well as chronic injuries or those with insidious or unknown onset, are not included.

[a] The NFL season structure changed in 2021 and 2022 to include 3 preseason games and 17 regular season games. Prior to 2021, the NFL season included 4 preseason games and 16 regular season games.

2.3. Statistical analysis

Descriptive statistics were computed for concussions and penalties between 2015 and 2022, including means and standard deviations. Incidence rates of concussions were calculated as the number of concussion events per total penalty called for 'lowering the helmet to initiate contact' and the number of concussion events per total play in each season. Descriptive statistics were analyzed using the Friedman test to assess overall differences between years. Subsequent pairwise comparisons were performed using the Wilcoxon signed-rank test with a Bonferroni correction to adjust for multiple comparisons. The trend analysis of concussion rates over the study period was performed using a linear regression model. The independent variable was the year, and the dependent variable was concussion rates. The relationship of concussion rates per year was assessed using Pearson's correlation coefficient. A p-value of less than 0.05 was considered statistically significant. Statistical analyses were conducted using SPSS software (IBM SPSS Statistics, Version 26).

3. Results

Between 2015 and 2022, there was a mean of 67 concussions per preseason and 159 per regular season (Table 1). The Friedman test indicated significant differences in injury trends across the years ($\chi^2 =$

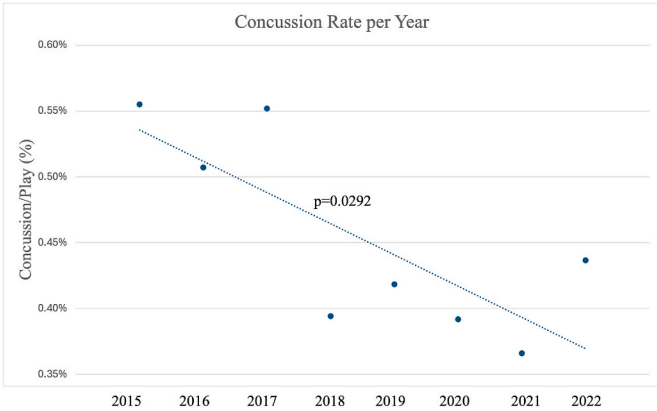

Fig. 1. Linear Regression Analysis of Concussion Rates per Year: Trendline indicates a statistically significant decrease in concussion rates from 2015 to 2022 ($p < 0.05$).

Table 3
Number of Concussions and 'Lowering the Helmet to Initiate Contact' Penalties per Year: Concussions reported from plays with 'Lowering the Helmet to Initiate Contact' penalty from 2018 to 2020 and incidence of concussions per penalty.[7,8].

Year	Number of Concussions	Total Number of Penalties	Concussions/Penalty (%)
2018	1	16	6.25%
2019	1	30	3.33%
2020	1	40	2.50%

16.0, $p = 0.0003$). Pairwise analysis utilizing the Wilcoxon signed-rank test and adjusted for multiple comparisons with a Bonferroni correction ($\alpha = 0.05/28 \approx 0.0018$) showed no statistically significant differences between years across the preseason, regular season, and overall totals. Each comparison between years resulted in non-significant findings, indicating that while overall trends over the years show variability, the specific year-to-year changes are not statistically significant.

The average incidence of concussions per play was 0.45% (Table 2). Concussion rates as a percent incidence of concussions per play showed a statistically significant negative trend ($R^2 = 0.575$, p-value = 0.0292), indicating a decrease in the rate of concussions from 2015 to 2022 (Fig. 1). A weak negative, but non-significant correlation was also observed between total plays and concussion rates ($R^2 = -0.331$).

The 'lowering the helmet' penalty was initiated for the 2018 season but was grouped under the unnecessary roughness penalty during the 2021 season. Thus, further data on this penalty was unable to be gathered. Between 2018 and 2020, 86 penalties for 'lowering the head to initiate contact' were called (Table 3). Overall, there was a 3.5% incidence of concussion with a 'lowering the helmet to initiate contact' penalty. Of the affected, all were offensive players. All lowering-the-helmet penalties that resulted in a concussion were from a defensive back, defensive tackle, or linebacker attacking a member of the offense. In 2020, one additional defensive player who was penalized left the game for concern for concussion but never officially entered the concussion protocol.

4. Discussion

The results from this study showed that there was an observed concussion incidence of 3.5% with 'initiating contact by lowering the helmet penalty. The incidence of concussion with 'lowering the helmet' penalties is greater than the overall incidence of concussions in the NFL per play in the regular season of 0.45%. Thus, these results show that the 'lowering the helmet' penalty addresses a movement with a considerably higher incidence of concussion than normal play. Despite the overall significant changes in the Friedman test, which suggests variability in injury occurrences over time, the lack of statistically significant differences between years with the Wilcoxon signed-rank test suggests fluctuations in concussion rates may be subtle and influenced by multiple factors, such as the addition of the 'lowering the helmet' penalty, while also increasing concussion education leading to players more often reporting concussion symptoms. The significant trend decrease in concussion rates per year, especially with a large decrease after the 2018 season, points to a possible positive impact of the 'lowering the helmet' penalty. The observed weak negative correlation between total plays and concussion rates indicates that although the number of plays per season has increased, there is not necessarily a proportional increase in concussions.

The "lowering the helmet to initiate contact" rule is defined as a foul if a player lowers his head to initiate contact and makes contact with the opponent with his helmet.[11] This penalty was initially implemented to prevent cervical spinal cord injuries after the devastating injury to the Pittsburg Steelers' Ryan Shazier in 2017.[12] However, this new penalty has also highlighted the effect that lowering the helmet to initiate contact can have on concussion occurrence. This study found that the NFL 'lowering the helmet' rule penalized a movement that has a higher incidence of concussion compared to the average play. Also, adding the rule could correlate with increased vigilance for concussions among players and the league. In addition, increased awareness following the 'lowering the helmet' rule has lowered the threshold to screen for the possibility of concussions. Despite only four total concussions during 'lowering the helmet' plays over the three years, direct helmet-to-helmet contact is likely to increase the risk of concussions in the NFL. Between 2017 and 2018, there was a 24% drop in reported concussions during the 2018 season, with the number between 2018 and 2019 for the number of concussions staying between 214 and 224, respectively.[13] Per the 2019 NFL Injury Takeaways, there have been more self-reported cases of players who have come forth to enter the concussion protocol, taking the first step in preventing long-term injury.[6] Before the 2022 preseason, the NFL required using the Guardian Cap, a soft helmet cover made of urethane, for certain positions during preseason training. The reported concussions during the 2022 preseason were at a seven-year low.[5] The NFL reported a 46% reduction in concussions for positions required to wear the Guardian Cap. However, there was a 50% increase in concussions for positions not required to wear the helmet.[5]

During the 2022 season, there was an 18% increase in concussions compared to the three-year average between 2018 and 2020.[10] During the 2022 End-of-Season Health & Safety Briefing, Dr. Allen Stills, NFL chief medical officer, attributed this change to a broadened concussion protocol that was initiated after Tua Tagovailoa of the Miami Dolphins suffered a string of concussions during the 2022 season and not due to an actual increase in concussions.[4] During the 2022 season, there was an average of 1.6 medical evaluations per game and an almost 200% increase in medical treatments on the field.[4]

There has been discussion that the NFL's "lowering the helmet" rule is more stringent than the NCAA's "targeting" rule, which was made to safeguard collegiate athletes subject to unnecessary roughness while defenseless. However, since the implementation of the "lowering the helmet" rule, there has been a decrease in concussion incidence between 2017 and 2020. Determining a causative relationship between the creation of the "lowering the helmet" rule and concussion incidence is challenging due to changes and improvements by the NFL in equipment, practice techniques, and overall culture perception toward concussions in the same period. However, it is encouraging that the concussion rate has decreased in the NFL post-initiation of rule changes, helmet safety, and player concussion education.[10]

A similar study in NCAA football that looked at the incidence of concussions secondary to targeting plays found that between the top two divisions, Division 1 Football Bowl Subdivision (FBS) and Division I Football Championship Division (FCS), there were 525 targeting

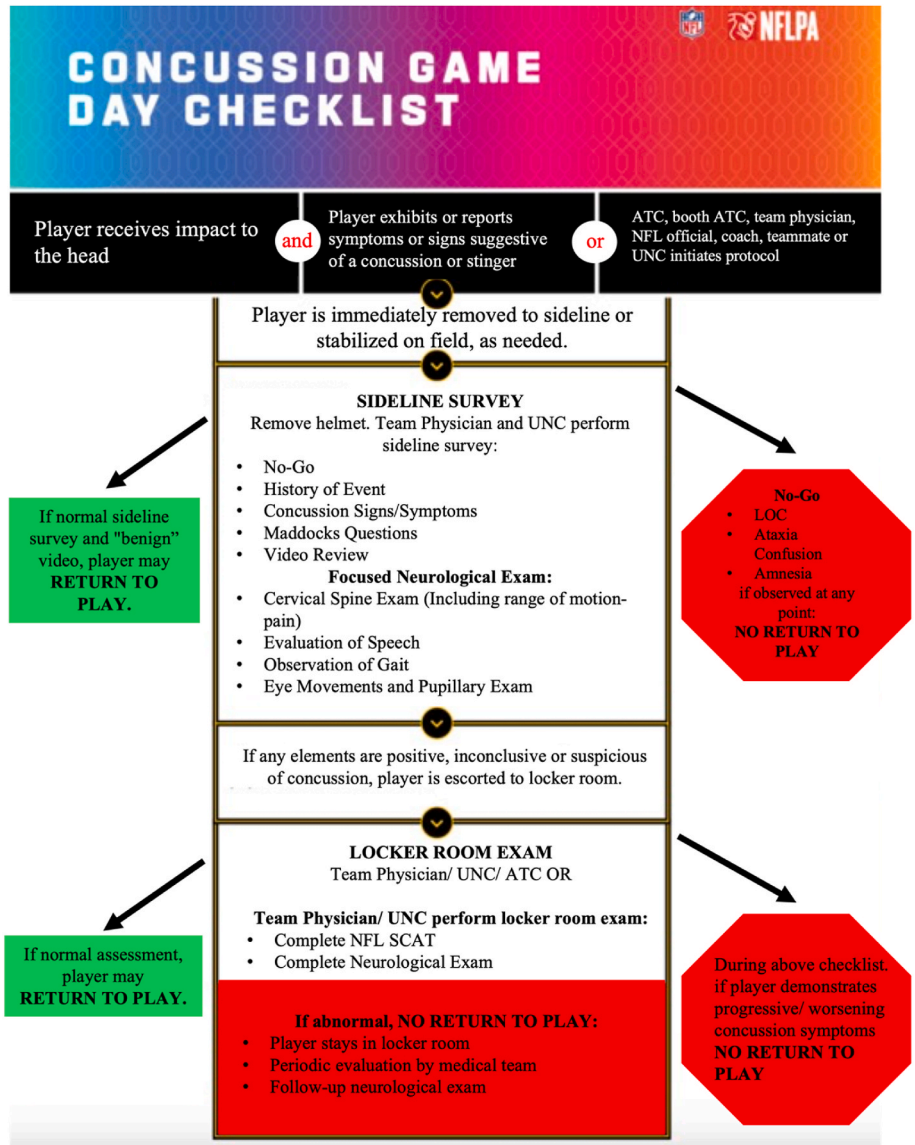

Fig. 2. NFL Concussion Game Day Checklist: NFL concussion game day checklist put forth by the NFL commission.[8].

penalties in the Power 5 athletic conferences. Translating this across the many different FBS teams conferences and teams, Aukerman et al. estimated 387 concussions that occurred during the study period between 2016 and 2019 across Division 1 football.[14] This resulted in a concussion incidence of 10% with targeting penalties, which is three times greater than the reported incidence of 3.5% with "lowering the helmet" penalties.

Another study in the National Football League examined the 'lowering the helmet' rule the year after its implementation. The 'lowering the helmet' rule significantly reduced the rate of concussions sustained by NFL players in the following regular season games, and games missed due to injury. The attribution was made that since the implementation of this rule, there has been a 40% decrease in SRC and no significance in the number of games missed due to targeting.[15]

Using tools such as the concussion game day checklist put forth by the NFL Commission (Fig. 2), there are measures already in place to identify concussions.[8] Considering that athletes do not report many concussions, there needs to be a thorough evaluation by medical staff, irrespective of whether a foul is called.[16] In addition, due to how fouls are recorded, players who commit repeat fouls can be identified, and coaching interventions for safer tackling decrease the risk of head trauma and ensure player safety in the future.

The clinical relevancy of this study is that when players position themselves to receive "lowering the helmet" penalties, whether in practice or games, there is a high risk of concussion. The incidence of concussion is Compared to the targeting rule, which helps protect defenseless offensive players, the "lowering the helmet" penalty protects both defensive and offensive players. Understanding the concussion risk associated with a player puts them in a "lowering the helmet" position, which will cause coaches and providers to teach athletes to avoid these positions at earlier levels of play, decreasing players' concussion risk as they progress to higher levels of play associated with more severe impacts due to increasing player size and strength.

5. Limitations

The study used data from the NFL Injury Data portfolio online and has a limited sample size due to the rule implementation for "Lowering the helmet" only being as recent as 2018. Therefore, it is difficult to retrospectively identify how many more concussions would have been identified if this rule had been adopted earlier. Additionally, following the 2020 season, the penalty for lowering the helmet to initiate contact

was grouped under the "unnecessary roughness" penalty. Even though referees call the penalty under the same guidelines, separating it from other unnecessary roughness penalties is impossible. The small sample size limited the analyses that could be conducted on 'lowering the helmet' penalties, but an analysis of the total concussions per year could be performed. Another confounding variable is that the data was gathered retrospectively, causing data not to be blinded to the researchers.

Furthermore, the players have an inherent bias to continue to play, driven by both financial and personal competitive pressures. Although this economic and social culture has changed over the years, it can still be a barrier for players to seek attention for concussion-related symptoms in fear of loss of play time, job/bonuses, or letting a team down. These factors could result in underreporting actual concussion events, which would be misrepresented in our data.

6. Conclusion

In conclusion, our results suggest a higher incidence of reported concussions following 'lowering the helmet to initiate contact' penalties compared to the overall incidence per play. While there was no significant change in concussions reported per year, there was a significantly decreasing trend in concussion rate per year, especially after the initiation of the 'lowering the helmet' penalty. While many factors can affect the true number of concussions, and it is difficult to ascertain the direct and indirect effects of the 'lowering the helmet' penalty, the results from this study point to this penalty serving as a negative reinforcement to styles of play that may increase the risk of injury. Further study into the effects of game alteration, such as penalizing tackling form as described in this study, could benefit how athletes of all levels train to prevent concussions in the long run.

Confirmation of ethical compliance

Because the study was a descriptive epidemiological study, no ethics committee approval was needed. Otherwise, the authors agree that all data gathering and reporting were done ethically without conflicts of interest or conscious bias.

Guardian/patient's consent

Due to the nature of this study being a descriptive epidemiological study, there was no need for any consent to be obtained from.

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CRediT authorship contribution statement

Garrett Waller: Methodology, Investigation, Data curation. **Grayson A. Domingue:** Investigation, Data curation, Writing – original draft. **Jaswanth Raj Kintada:** Writing – original draft. **McKenna W. Box:** Visualization, Data curation. **Garrett Steinmetz:** Methodology, Data curation. **Douglas Rowles:** Supervision.

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